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In [1]: import matplotlib.pyplot as plt
import numpy as np
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In [2]: def lin_interpolate(x0,x1,y0,y1,x):
return y0 + ((y1-y0)/(x1-x0))*(x-x0)
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In [3]: def quad_interpolate(x0,x1,x2,y0,y1,y2,x):
return y0*((x-x1)*(x-x2))/((x0-x1)*(x0-x2)) + y1*((x-x0)*(x-x2))/((x1-x0)*(x1-x2)) + y2*((x-x0)*(x-x1))/((x2-x0)*(x2-x1))
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In [4]: def poly_n_interpolate(xn,yn,x):
pn = 0
n = len(xn)

for i in range(n):
    l_i = 1
    for j in range(n):
        if i != j:
            l_i *= (x-xn[j])/(xn[i]-xn[j])
    pn += yn[i]*l_i

return pn
```

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In [5]: def newton_interpolate(xn,yn,x):
n = len(xn)
slopes = yn.copy()

for i in range(1,n):
    for j in range(n-1,i-1,-1):
        slopes[j] = (slopes[j]-slopes[j-1])/(xn[j]-xn[j-1])

pn = slopes[0]
term = 1
for i in range(1,n):
    term *= x-xn[i-1]
    pn += slopes[i]*term

return pn
```

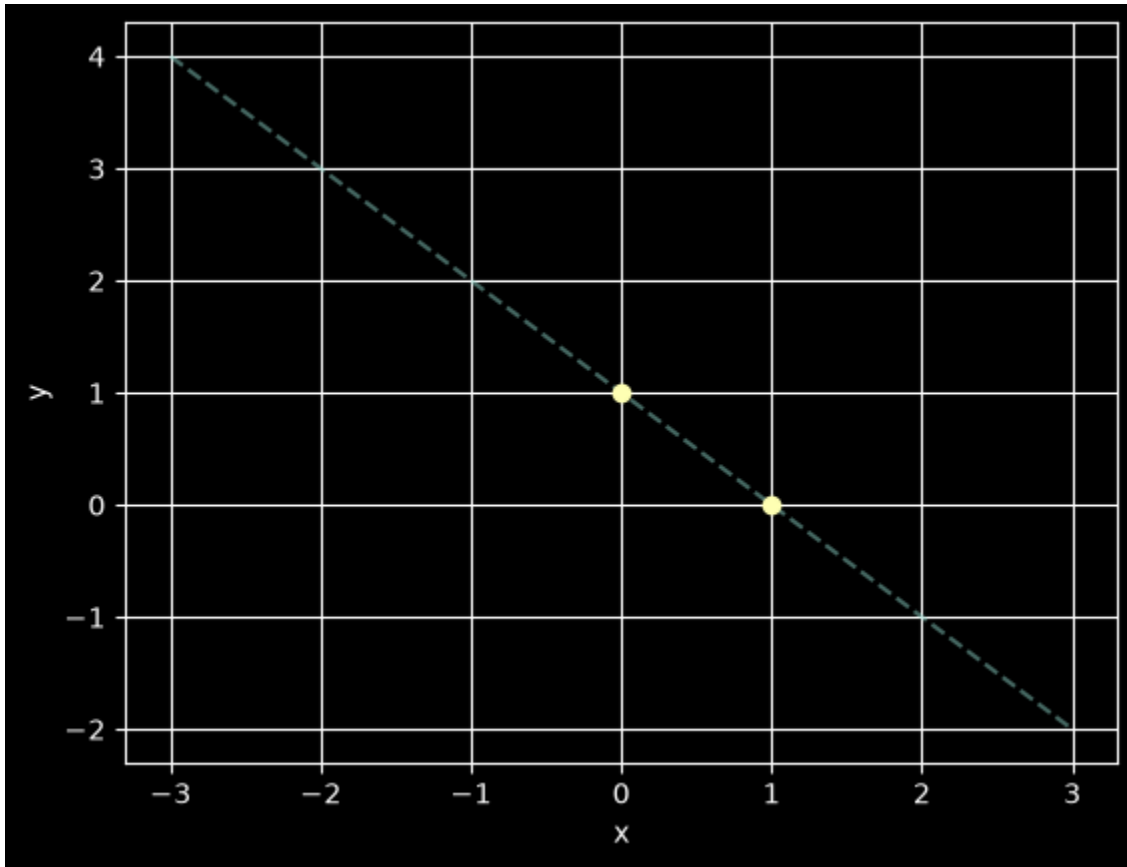
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In [6]: x0 = 0
y0 = 1
x1 = 1
y1 = 0

x_range = np.linspace(-3,3,100)
y_range = []
for i in x_range:
    y_range.append(lin_interpolate(x0,x1,y0,y1,i))

plt.plot(x_range, y_range, linestyle='--', alpha=0.5)
plt.plot([x0,x1],[y0,y1], marker='o', linestyle='None')
plt.xlabel('x')
plt.ylabel('y')
```

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plt.grid(True)
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```
plt.show()
```



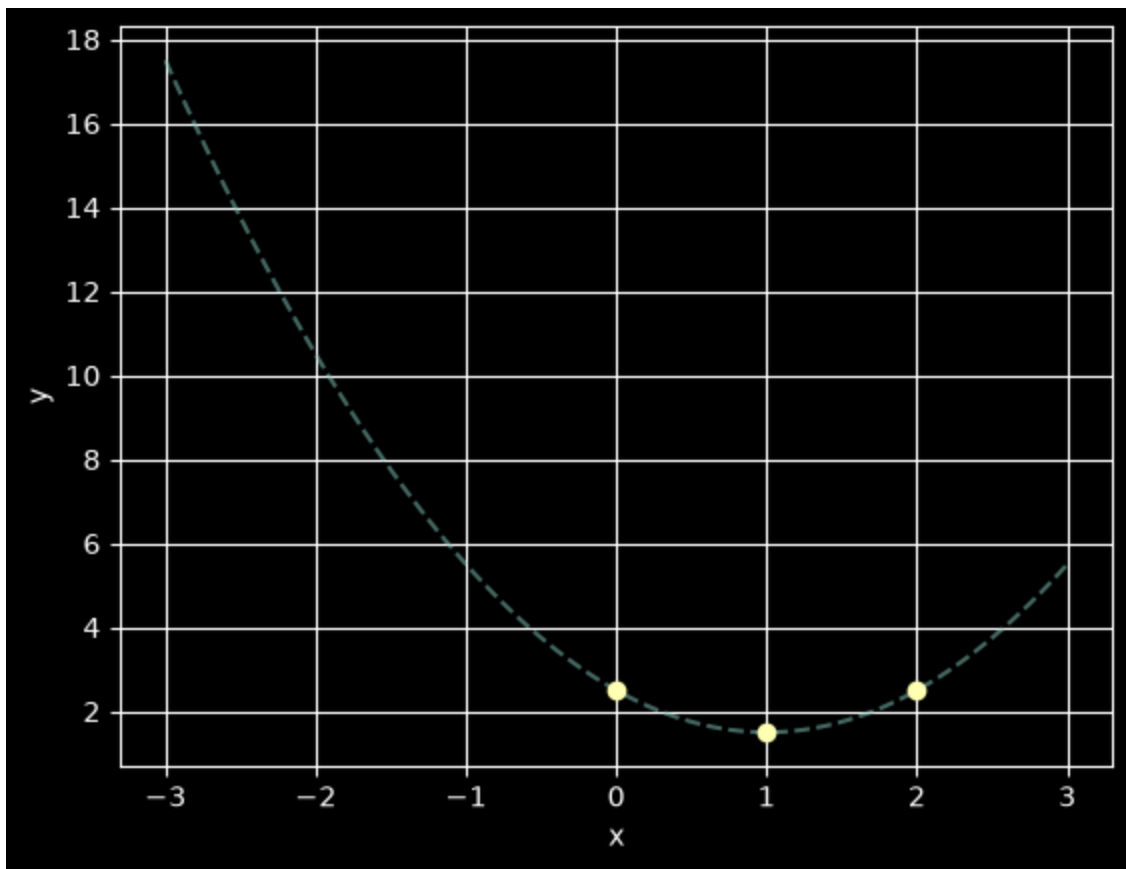
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In [7]: x0 = 0
y0 = 2.5
x1 = 1
y1 = 1.5
x2 = 2
y2 = 2.5

x_range = np.linspace(-3,3,100)
y_range = []
for i in x_range:
    y_range.append(quad_interpolate(x0,x1,x2,y0,y1,y2,i))

plt.plot(x_range, y_range, linestyle='--', alpha=0.5)
plt.plot([x0,x1,x2],[y0,y1,y2], marker='o', linestyle='None')
plt.xlabel('x')
plt.ylabel('y')

plt.grid(True)

plt.show()
```



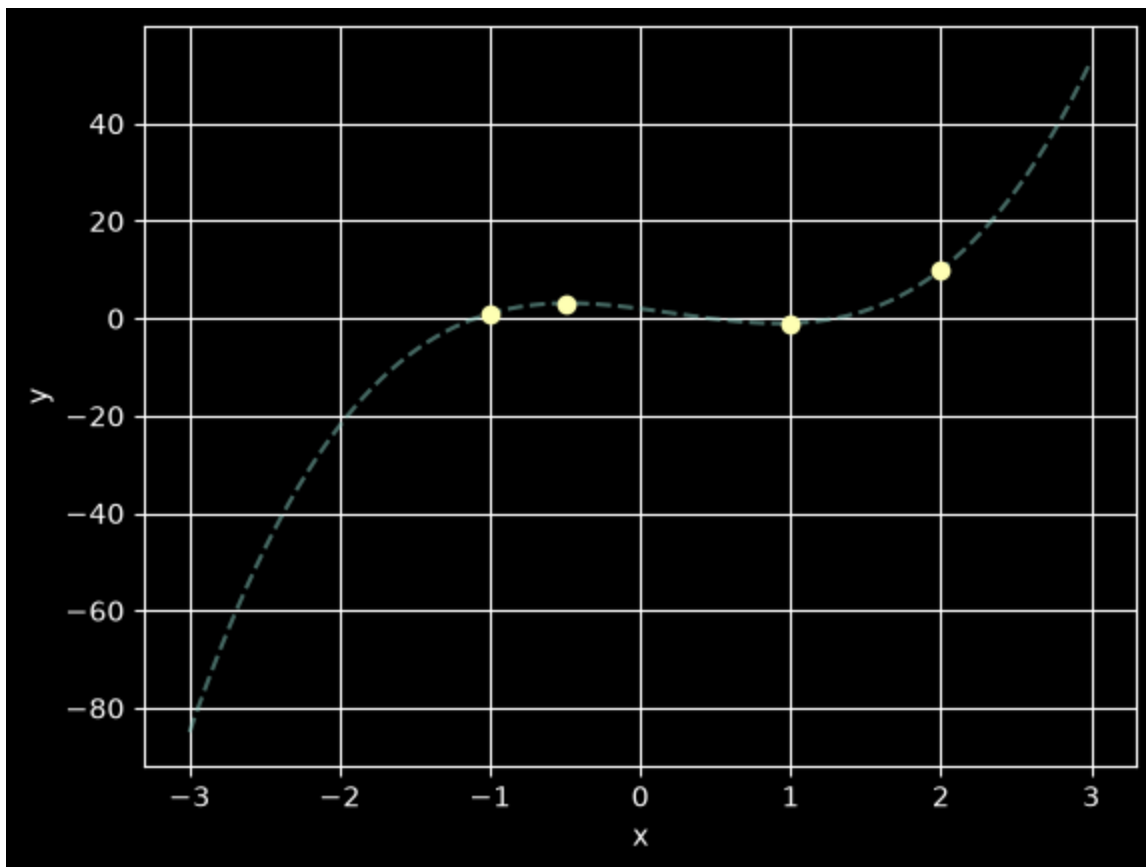
```
In [8]: xn = [-1,-0.5,1,2]
        yn = [1,3.125,-1,10]

        x_range = np.linspace(-3,3,100)
        y_range = []
        for i in x_range:
            y_range.append(poly_n_interpolate(xn,yn,i))

        plt.plot(x_range, y_range, linestyle='--', alpha=0.5)
        plt.plot(xn, yn, marker='o', linestyle='None')
        plt.xlabel('x')
        plt.ylabel('y')

        plt.grid(True)

        plt.show()
```



```
In [9]: xn = [-1,-0.5,1,2]
        yn = [1,3.125,-1,10]

        x_range = np.linspace(-3,3,100)
        y_range = []
        for i in x_range:
            y_range.append(newton_interpolate(xn,yn,i))

        plt.plot(x_range, y_range, linestyle='--', alpha=0.5)
        plt.plot(xn, yn, marker='o', linestyle='None')
        plt.xlabel('x')
        plt.ylabel('y')

        plt.grid(True)

        plt.show()
```

